PA194NS: a different 194-state PA consistency machine

A corpus comparison, a changed command protocol, and an explicit proof-to-computation compiler

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Abstract

No machine with fewer than 194 working states was established. We supply the requested alternative: a materially different 194-state, binary, single-tape, blank-start Turing machine, **PA194NS**, together with an ordinary mathematical proof of

$$\mathrm{PA} \vdash \big(\neg\,\mathrm{Halt}(\mathrm{PA194NS}, \mathrm{blank}) \leftrightarrow \mathrm{Con}(\mathrm{PA})\big).$$

The new machine combines no-push modus ponens returning its conclusion to the current-formula register with a symmetric Shift implementation that tests only the cached formula. These features occur separately in the supplied designs; their literal composition and its complete command compiler are the new result. The combination needs a new contradiction detector: a plain Shift can discard a false current formula.

The construction has 3,650 primitive source sites and 4,750 source outcomes. Two symbolic physical checks cover arbitrary natural register values, both decrement outcomes, a variable header gap, and an arbitrary nonempty spare list. Exact blank execution reaches its complete source frame after 57,768 transitions. Eight simultaneous folds have 32 explicit local contracts. The table differs from every recognized existing 194-state graph, even allowing arbitrary control renaming, a changed spanning start, tape reflection, and global symbol exchange.

The comparison is pinned to ProveIt commit `1e407ae841e4666f932777fad977ba784ebbcfd5`. It covers 46 research packets and recognizes 3,741 literal-table occurrences, 666 byte identities, and 476 action-graph classes. The 28 old 194-state classes include supplementary and explicitly rejected machines, not 28 independently proved consistency results. The logical PA calibration is inherited and identified precisely. Neither a newly compiled end-to-end Lean/Rocq theorem nor a completed iPRA endpoint is claimed.

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1 Result, conventions, and exact identity

The principal result is the table `machine/machine.tm`. Its working controls are $0, \ldots, 193$, its initial control is 0, and its separate terminal label is 194. The tape is two-way infinite, with alphabet $\{0, 1\}$, initially zero everywhere; the initial head coordinate is zero. Every transition writes one bit and moves one cell left or right. There are exactly 388 rows, two per working control, and no outgoing halt row. All 194 working controls are reachable in the directed transition graph from control zero. This graph fact does not assert that the actual blank execution visits every control.

The machine format is

```
state  read  write  L-or-R  target.
```

Appendix A prints the complete table. The exact SHA-256 of the dense file is

`19935635f2c5722f5a72977d20a7019764a28573e12d1d0bce398245576c33b1`.

The native presentation uses start 74 and halt 254, with SHA-256

`0b72fc7decfe10ffe30c967f59ddfc570bc7f92177f67b72735b82a0d2fe9fb8`.

The array `machine/labels.json` is the dense-to-native bijection, including the halt label. Every row is checked under this bijection.

Let $H_{M_{\mathrm{NS}}}$ denote the ordinary arithmetic sentence that a finite literal tape trace starts at this blank configuration and ends in halt. Use a fixed ordinary first-order PA proof system for $\mathrm{Con(PA)} = \neg\exists p\ \mathrm{Prf}_{\mathrm{PA}}(p, \ulcorner 0 = 1 \urcorner)$. The theorem concerns these coded finite witnesses inside PA:

$$\mathrm{PA} \vdash H_{M_{\mathrm{NS}}} \leftrightarrow \exists p\ \mathrm{Prf}_{\mathrm{PA}}(p, \ulcorner 0 = 1 \urcorner). \tag{1}$$

Negation gives the desired consistency biconditional. The argument uses syntactic proof transformations, loop induction, and finite-trace composition; it does not assume PA's consistency within PA. Actual nonhalting in the standard model follows if PA is consistent.

1.1 What is new and what is inherited

Relative to the supplied PA194CS table, precisely one native row changes:

$$(164, 1): \quad 1R130 \longmapsto 1R114. \tag{2}$$

This redirects the successful modus-ponens return to a different existing register-copy routine. The three-site transfer to the cache becomes a seven-site copy to the current-formula register while retaining the numerical offset. Both the resulting command semantics and the proof compiler change. The arithmetic tape routines, separate-head axiom calculus, axiom-emission order, scheduler, and 67-reader AX grammar are retained.

The count is reconstructed from the unfurled table, rather than inferred from its name:

$$\underbrace{135}_{\text{fixed controls}} + \underbrace{67}_{\text{AX readers}} - \underbrace{7}_{\text{AX folds}} - \underbrace{1}_{\text{controller fold}} = 194. \tag{3}$$

The eight identifications are checked simultaneously. The controller fold $41 \mapsto 330$ is inherited from PA194CS; it is not an eighth new AX opportunity. No existing state saving is counted twice.

The no-push command and symmetric comparison are credited to their supplied PA and ZFC sources [3, 4, 7]. The contribution here is their new literal composition, its exact source binding, independent physical audit, all-store proof compiler, and comparison against the current corpus. The claim is novelty in this pinned corpus, not an assertion that the general programming ideas have never appeared elsewhere.

2 Comparison of all supplied research packets

2.1 Scope and reproducible census

The snapshot contains 41 filed research packets under `Computability/BusyBeaver/Research/PA/docs` and five incoming ZIP packets. Both locations are included, along with literal tables elsewhere in the PA tree and recursively nested ZIPs. The complete 46-entry ledger is in Appendix C and `comparison/packet_ledger.csv`. The imported James Chang PA196 and PA209 developments are also consulted for the underlying source and the formal proof boundary.

The recognizer accepts complete numeric five-column tables in `.tm`, `.txt`, `.tsv`, and `.csv` files, with binary read/write symbols, L/R movement, and one undefined terminal target. It recognizes literal machines, not arbitrary tables embedded only in Lean, JSON, Python, or prose. Source-only proposals are still discussed in the packet ledger; the numeric census is explicitly narrower. Duplicate ZIP payloads are traversed once, with every encountered archive reference retained. No archive or table-read failure was recorded.

Quantity	Pinned-snapshot result
Filed / incoming research packets	41 / 5
Recognized literal-table occurrences	3,741
Distinct literal byte strings	666
Action-graph classes with a spanning root	476
194-state byte strings / action graphs	51 / 28
195-state byte strings / action graphs	26 / 16
196-state byte strings / action graphs	163 / 99
Distinct nested-or-top-level ZIP payloads / references	19 / 31
Read errors	0

Two disconnected synthetic tables, of sizes six and nine, have no spanning root and are excluded from the graph-class count. The recognized corpus contains no table of size 190–193. These are inventory facts, not state lower bounds and not counts of proved consistency machines. Older packets report larger occurrence counts because their snapshots contained more duplicate archives; those numbers are not interchangeable with the current census.

For a graph comparison, every root that reaches the entire working graph is used for a deterministic breadth-first serialization. Read labels, write bits, movement directions, and halt edges are retained. Minimizing over these exact serializations removes dependence on control names and on a chosen spanning start. The new-candidate test additionally allows tape reflection and global read/write complementation. All 776 transformed roots of PA194NS were tested against the 51 old 194-state byte forms, using exact tuples as well as recording their hashes. No equality occurs.

2.2 The 194-state frontier

The 28 recognized graph classes comprise 16 principal reported designs, six supplementary designs, three research seeds, and three explicitly rejected experiments. "Principal" describes the packet's role and evidence; it does not claim that every historical proof has been rerun independently in this work. Appendix B and `comparison/frontier194.csv` identify every class and its exact bytes.

Family	Design distinction and implication for further reduction
PA194-67; PA194-68	Same separate-head source, respectively 67 readers with seven folds and 68 readers with eight. Both leave 60 effective AX controls; their fold counts cannot be added.
PA194-AD	Physical A/D register exchange. This changes the machine graph and startup duration, but preserves the logical source.
PA194C; PA194CS	Cached commands, then symmetric destructive comparison. Five packets repeat the same PA194C dense table. PA194CS exchanges a source saving for a controller reader and recovers that reader by a separate fold.
PA194N	Successful MP returns its conclusion to the top without pushing a zero, while Shift still tests $t + h = 1$. Its proof compiler supplies the missing spare zero.
PA194Q; PA194D	Q exchanges the premises of the quantifier-distribution head. D emits $\Gamma(D)$ instead of $\Gamma(A)$ and has 69 readers with nine folds. These change numerical axiom production and require selector/proof arguments.
PA194J; A; F; CSZ/X	Replace jumps by decrements of registers proved zero at those sites. J combines ten replacements; A changes an entry jump; F changes the final emitter return. CSZ and X are exactly the same dense table.
PA194R	F's source graph with a new AX association: 1,111 source-address paths change while the resolved instruction graph is retained.
M4R; M5; M6	New generic pairing/radix allocation. M5 and M6 replace the formula coefficient four by five and six, with proof-code recoding; all have 194 states.
PA194NS	**New: no-push MP-to-top combined with cache-only symmetric Shift. The proof compiler uses an explicit detector because a plain Shift can lose the contradiction.**

Two especially relevant rejected ideas illustrate separate obligations. A shortened return scanner can misread a legal source address as the separator even when local reader folds pass. A known-zero source change can satisfy all source contracts yet make the physical blank initializer halt after 1,361 steps. A source invariant and a complete blank-to-source binding are therefore both required. Neither kind of rejected table is counted as a successful consistency construction.

2.3 The larger lineages

The 195-state frontier includes the bundled-diagonal PA195, separate-head 195S, cached 195C, diagonal-cached PA195DC, intermediate reader allocations, and the M195S8 scanner variant. Their differences occur at three levels: which numerical heads are emitted, where command results and spare zeros are stored, and how the source tree is physically decoded. The original PA195 and separate 195S allocations were $134 + 68 - 7$ and $134 + 67 - 6$; the primary 194 layout combines the smaller reader allocation with the larger compatible fold set.

The PA196 work explores reader reassociation, helper-return changes, register placement, cached commands, known-zero jumps, and changes to the separation/diagonal heads. The NEG18 proposal is a useful counterexample to optimizing the wrong quantity: its shorter encoded negation yields a 197-state implementation, despite removing source instructions. Source length and the number of binary TM controls are different costs.

The older U208/U209 family develops entry-context and history-sensitive quotients, alternate helpers, unary power-cover sources, and explicit counterexamples to further local contractions. Several reports give different explanations of the same quotient. Their actual literal bytes and proof obligations matter more than archive names. James Chang's imported PA209 and PA196 packages provide a distinct development of the same broad arithmetic proof-search strategy. The 46-packet ledger records the individual contributions and unsuccessful directions rather than presenting all copied tables as independent improvements.

2.4 What the proof-assistant work currently establishes

The pinned PA README explicitly distinguishes the machine-specific consistency reductions from ordinary finite-run theorems, Python certificates, and conditional arithmetic endpoints [1]. It states that neither machine-specific direction of the desired iPRA equivalence is complete for the primary PA194, original PA195, or PA196. The imported PA196 result supplies a PA implication from literal nonhalting to PA consistency; this is not by itself the reverse direction.

The supplied later manuscripts give an ordinary syntactic PA calibration for the changed numerical calculus [2, 5, 6]. We retain and use those arguments, including their uniform PA-proof-producing soundness construction. The new machine is proved by composing that identified mathematical dependency with the new command compiler and physical refinement below. No conditional Lean endpoint is silently treated as a proved biconditional.

3 PA194NS: crossed command semantics and a complete proof compiler

PA194NS combines the no-push, top-result modus-ponens command of PA194N with the cache-only symmetric Shift implementation of PA194CS. The underlying numerical axiom calculus and its ordered emitter are unchanged. The combination is not an immediate source equivalence with either parent: the proof-command compiler must be checked again, because its old final contradiction detector no longer works.

The machine has 194 working states, 388 rows, start state 0, halt label 194, a two-way binary tape initially zero, and only left/right moves. In native labels, the entire edit from PA194CS is

$$(164, 1): \quad 1R130 \longmapsto 1R114.$$

The literal state count remains

$$135 \text{ fixed controls} + 67 \text{ AX readers} - 7 \text{ AX folds} - 1 \text{ controller fold} = 194.$$

The seven AX folds and the controller identification $41 \mapsto 330$ are verified against the final table simultaneously, with 32 local cases. The source now has 3,650 primitive sites and 4,750 branch outcomes.

3.1 Exact command semantics

Write

$$p(a,b) = \frac{(a+b)(a+b+1)}{2} + b, \qquad [a;w] = p(a,w), \qquad I(a,b) = 4p(a,b), \qquad U(i,a) = 4p(a,i) + 2.$$

A numerical stack has the literal identities

$$\mathrm{hd}[a;w]=a,\quad \mathrm{tl}[a;w]=w,\quad [0;0]=0,\quad \mathrm{hd}(0)=\mathrm{tl}(0)=0.$$

Thus the notation does not assume that numerical stacks form a free-list algebra. Code zero is truth, code one is the designated contradiction, and $I(0,0)=0$ is an intentional constructor alias.

A command store is (t,w,d,h): current top, Cantor stack, numerical offset, and cached formula. For $b=d+x$, the new MP and Shift commands are

$$\mathsf{MP}_x(t,w,d,h)=\begin{cases}(b,w,b,0), & I(t,b)=h,\\ (0,w,b,0), & I(t,b)\neq h,\end{cases} \tag{4}$$

$$S(t,w,d,h)=\begin{cases}\mathsf{halt}, & h=1,\\ (h,\mathrm{tl}(w),d,\mathrm{hd}(w)), & h\neq 1.\end{cases} \tag{5}$$

The AX command is the unchanged separate-head emitter. For $D=d+x$ and $H=h+y$,

$$\mathsf{AX}_{A,x,y}(t,w,d,h)=(g(H,D,A),[q_1,\dots,q_9,t;w],0,0). \tag{6}$$

Here $q_1=\Gamma(A)=I(A,U(A,A))$, $q_2,\dots,q_7$ are the six inherited PA196 heads in reverse emission order, and

$$q_8=\rho(D,A,H,d)=I(I(I(H,d),D),I(I(D,H),I(A,H))),\qquad q_9=U(d,h).$$

The top g is the existing guarded separation gate. The executable free-constructor audit checks every complete formula tree, parameter, stack position, and guard suffix from the delivered grammar. All $q_1,\dots,q_8,g$ have residue zero modulo four; q_9 has residue two and is at least two. None is code one, for any natural input store. This arithmetic fact requires no consistency assumption.

These semantics distinguish PA194NS from both parents. From $(t,w,d,h)=(1,0,0,0)$, the PA194N Shift halts, whereas the PA194NS Shift returns $(0,0,0,0)$. Also, $p(0,1)=2$, so $I(0,1)=8$; $\mathsf{MP}_1(0,0,0,8)$ is $(1,0,1,0)$ in PA194NS and $(0,0,0,1)$ in PA194CS. A successful implication therefore lands in a different register from the register tested for contradiction. The next plain Shift can discard a false current top; the compiler below uses an explicit detector that repairs this mismatch.

3.2 Binding the new source block

In the PA194CS source, the success block has exactly the three sites 3634–3636, at binary prefix `1001010111`. It transfers D to H. PA194NS replaces that subtree by the following seven-site block:

3634	dec D [3638, 3635]
3635	inc $T \to$ 3636
3636	inc $X \to$ 3637
3637	jump 3634
3638	dec X [3641, 3639]
3639	inc $D \to$ 3640
3640	jump 3638.

At successful MP entry, the unchanged preceding constructor/comparison code supplies $(T,H,X,D)=(0,0,0,b)$. The first loop preserves $T+D=b$ and $X=T$, and ends with $(T,X,D)=(b,b,0)$. The second preserves $D+X=b$ and ends with $(T,X,D)=(b,0,b)$. Both loops terminate by induction on their decremented register, preserve W and every other register, and perform no stack push. This

proves the success clause of (4); the failure path is unchanged. The exact source comparison checks every outside instruction after index relocation and proves that the only outside entrance to the replacement is the zero exit from site 3628 to site 3634. No outside edge enters its interior.

The cached symmetric Shift block is retained byte for byte from PA194CS. It clears T, transfers H into T and X, increments the now-zero H, and compares $(H, X) = (1, h)$. The inherited destructive comparison halts exactly on equality and otherwise clears both operands before decoding one stack cell. Induction on the sum of its two inputs proves these properties. Thus it realizes (5) under the common command-boundary invariant $X = 0$. With arbitrary incoming scratch x, the inherited comparison of the restored and symmetric tails instead tests $h + x = 1$ in both implementations. That stronger tail-equivalence statement is not being used to assert the command interface for arbitrary scratch.

3.3 Two AX identities and the changed Pop contract

Let $A_0 = \mathsf{AX}_{0,0,0}$. Directly from (6) and (5),

$$\mathsf{AX}_{A,x,y}; S^9(t, w, d, h) = (q_8, [t; w], 0, U(d, h)), \tag{7}$$

$$\mathsf{AX}_{A,x,y}; S^{10}(t, w, d, h) = (U(d, h), w, 0, t). \tag{8}$$

Every displayed Shift is safe, since its tested cache is zero or one of the generated non-one codes. In particular these are total identities for arbitrary natural t, w, d, h, including $t = 1$.

Define

$$\mathsf{Push} = A_0; S^9; \mathsf{MP}_0, \qquad \mathsf{Pop} = A_0; S^{11}, \qquad \mathsf{Discard} = \mathsf{MP}_0.$$

Push is total and obeys

$$\mathsf{Push}(t, w, d, h) = (0, [t; w], 0, 0).$$

Indeed its final MP compares an implication of residue zero with the universal code of residue two in (7), and hence fails with offset zero. Also, both MP branches coincide when $d = x = 0$, giving

$$\mathsf{Discard}(t, w, 0, h) = (0, w, 0, 0)$$

for arbitrary old cache h.

The correct Pop contract is now a case distinction:

$$\mathsf{Pop}(t, [a; w], d, h) = \begin{cases} \mathsf{halt}, & t = 1, \\ (t, w, 0, a), & t \neq 1. \end{cases} \tag{9}$$

The first ten Shifts are (8); the eleventh tests exactly the saved old top t. If it passes, Cantor projection gives the stated store. This applies equally when $a = w = 0$, because $[0; 0] = 0$. For the zero stack, a nonfalse Pop returns $(t, 0, 0, 0)$; a zero Push onto the zero stack returns the zero stack. No fictitious sentinel cell is counted in either identity.

3.4 Axiom selection and generalization

For an arbitrary incoming $(t, w, 0, 0)$ and $1 \leq j \leq 7$, the word

$$\mathsf{Discard}; \mathsf{AX}_{A,D,H}; S^{j+1}; \mathsf{Pop}^{9-j}$$

returns $(q_j, w, 0, 0)$. The selected q_j is not one, so every subsequent Pop uses the second case of (9). The last Pop consumes the saved zero and clears the cache. For $u \leq a$, the dominated head is selected by

$$\mathsf{Discard}; \mathsf{MP}_u; \mathsf{AX}_{b,a-u,c}; S^9; \mathsf{Pop}.$$

This produces $(\rho(a,b,c,u),w,0,0)$. From the zero top/cache store, MP_u gives $(0,w,u,0)$ for every u: at $u=0$ the two branches agree, and at $u>0$ the implication code is positive and differs from the zero cache. A permitted gate uses

$$\mathsf{Discard};\mathsf{AX}_{A,D,H};\mathsf{Pop}^{10},$$

returning $(g(H,D,A),w,0,0)$. Truth uses Discard alone.

Generalization at variable label i is implemented by

$$\mathsf{Gen}_i=\mathsf{Push};\mathsf{MP}_i;S;\mathsf{AX}_{i,0,0};S^{10}.$$

From $(a,w,0,0)$, the first three pieces give $(0,w,i,a)$. Equation (8) then gives $(U(i,a),w,0,0)$. This word is total even when $a=1$: its first Shift tests zero, and all later Shifts test only generated non-one heads. No hidden assumption that an arbitrary formula is nonfalse occurs here.

3.5 Modus ponens and the reserved zero

The compilation invariant is: from every $(t,w,0,0)$, the word $C(p)$ for a numerical-calculus derivation p either halts or returns $(\mathrm{concl}(p),w,0,0)$. Axioms and generalization have just been handled. Suppose the premises are a and $I(a,b)$. In the next display, $C(a)$ and $C(I(a,b))$ mean compilation of the given premise derivations, not a search for proofs of those formulas. Use

$$\mathsf{Discard};\mathsf{Push};C(I(a,b));\mathsf{Push};C(a);\mathsf{Push};S;S;\mathsf{MP}_b;\mathsf{Pop}. \tag{10}$$

The initial Discard/Push reserves a zero below both recursive premise computations. If neither recursive word halts, the store just before the two Shifts is

$$(0,[a,I(a,b),0;w],0,0).$$

The first Shift returns $(0,[I(a,b),0;w],0,a)$. The second halts if $a=1$; otherwise it gives $(a,[0;w],0,I(a,b))$. MP succeeds and returns $(b,[0;w],b,0)$. The final Pop halts if $b=1$ and otherwise returns $(b,w,0,0)$. The offset is cleared and the reserved zero is consumed exactly once. This verifies every branch of the required halt-or-return contract. At $a=b=w=0$ the proof uses literal Cantor projections and $I(0,0)=0$; it does not appeal to distinct syntactic constructors.

Induction on the finite derivation establishes the compilation invariant. The recursive compiler is primitive recursive: each derivation rule is replaced by a fixed finite word together with recursive premise words and numerical parameters already present in the coded derivation. PA proves its correctness by induction on that coded derivation.

3.6 A detector valid from every carried store

Appending a plain Shift is insufficient: $(1,0,0,0)$ would continue to zero. Instead define

$$\mathsf{Detect}=A_0;S^{11}=\mathsf{Pop}.$$

By (9), Detect halts from $(1,w,d,h)$ for every natural w,d,h. Given a derivation p of code one, the concrete command word

$$A_0;C(p);\mathsf{Detect}$$

halts from *every* incoming store. The first AX clears offset and cache while leaving some top and stack; the compiler works with arbitrary such top and stack. It either halts already or returns current

top one, which Detect then detects. This argument does not assume that the scheduler resets its store between unsuccessful command words.

The unchanged scheduler has fresh counter initially two and a total Cantor command decoder whose positive continuation codes decrease. Let $E(\epsilon) = 0$ and $E(c :: w) = 1 + \text{encodeCommand}(c, E(w))$ be the positive continuation convention. The exact fresh-code forms are

$$\begin{aligned}\text{encodeCommand}(S, z) &= p(0, z),\\ \text{encodeCommand}(\mathsf{MP}_x, z) &= p(1, p(x, z)),\\ \text{encodeCommand}(\mathsf{AX}_{A,x,y}, z) &= p(A + 2, p(y, p(x, z))).\end{aligned}$$

A fresh candidate uses its raw code directly. A positive continuation n first decodes raw $n - 1$, whose residual is at most $n - 1 < n$. Thus a fresh candidate n takes at most $n + 1$ decoded commands before its continuation is exhausted, unless it halts earlier. If $z = E(A_0; C(p); \mathsf{Detect})$, the fresh candidates

$$p(k + 2, p(0, p(0, z))) \qquad (k \in \mathbb{N})$$

form a strictly increasing, unbounded family. Their decoded words prepend one arbitrary AX to the all-store halting word. Thus omitting the finitely many candidates below two causes no loss. PA verifies termination of each command macro and decoding loop by the inherited natural-number ranking arguments. For a given proof, an explicit sufficiently large prefix code and the finite search up to it produce a finite halting source run. Evaluation of each candidate is a bounded traversal of at most $n + 1$ commands, each with a total primitive-recursive macro implementation; halted configurations can be treated as absorbing. A bounded outer iteration up to the chosen candidate code then constructs the complete finite source trace. Code construction and finite trace concatenation are primitive recursive, and PA proves their correctness by ordinary induction.

3.7 Soundness and the calibration dependency

For soundness, annotate the current top t, cached formula h, and every live entry in the canonical numerical stack with a proof in the retained calculus, or with the inherited PA proof of its arithmetic translation. For $w > 0$ its Cantor tail is strictly below w, so the canonical stack is obtained by a primitive-recursive bounded recursion. A popped zero stack supplies truth, and pushing zero onto zero introduces no live entry.

AX adds the same valid axiom heads, the generalization of the old cache, and the saved old top. MP success takes the annotated formulas t and $I(t, b) = h$ to the annotated conclusion b, retains the stack and annotates its zero cache by truth. MP failure outputs truth and retains the stack. The numerical offset is not required to be a theorem. Shift transports annotations; if it halts, its cache is code one and already has a proof. These are uniform syntactic proof transformations, so PA proves that a finite source halt yields a PA refutation. No reflection or uniform truth predicate for PA is used.

The inherited logical-calibration dependency is the conclusion-preserving, primitive-recursive translation between PA refutations and refutations in the separate-head numerical calculus. Its arithmetic interpretation uses finite binary membership and PA's finite-comprehension proof templates; its completeness side uses the established reverse membership rewriting and neighborhood-equality interfaces. The unchanged proof texts are retained under `references/logical`. The present change neither replaces those arguments by the repository's conditional iPRA endpoints nor claims a new kernel-checked proof of them.

Combining that dependency with the new all-store compiler and annotation argument gives, for the new source R_{NS},

$$\text{PA} \vdash \big(\text{Halt}(R_{NS}) \leftrightarrow \exists p\ \text{Prf}_{\text{PA}}(p, \ulcorner 0 = 1 \urcorner)\big).$$

3.8 From the source to the exact tape table

The complete 4,750 primitive outcomes are checked for arbitrary natural register values, both decrement branches, an arbitrary gap parameter and an arbitrary finite nonempty list of spare register values. An independent literal-tape checker reconstructs the source by the explicit address splice, compares two control-flow resolvers, rederives the four generic list lemmas from the actual transition rows, and checks the exact complete frames. Increments include their specified spare-list effects.

Literal blank execution reaches the complete source-25 frame in 57,768 steps, at native state 84 and head coordinate -2, with logical registers $(2, 0, 0, 0, 0, 0, 0, 0, 0, 0)$ and 26 spare registers of value one. Two different finite tape representations compare the entire tape, including its zero exterior; there is no earlier halt. The dense/native map conjugates every row exactly.

Every source step has a terminating positive-length physical simulation, with no premature physical halt. A finite halting source trace therefore produces a physical halting trace by finite concatenation after startup. Conversely, if the physical machine first halted after T steps while its first $T + 1$ source steps were all nonhalting, their positive simulation segments would concatenate into the same deterministic physical run for more than T nonhalting steps, a contradiction. This yields both finite-trace implications in PA. Consequently, for the exact delivered 194-state table,

$$\boxed{\mathrm{PA} \vdash (\neg\, \mathrm{Halt}(\mathrm{PA194NS}, \mathrm{blank}) \leftrightarrow \mathrm{Con}(\mathrm{PA})).}$$

The new result is an ordinary mathematical PA-provability argument with finite and symbolic executable certificates. These certificates are not a new end-to-end Lean/Rocq theorem; actual standard-model nonhalting is conditional on PA's consistency. No state minimality conclusion follows from the unsuccessful smaller-machine searches.

4 Verification evidence and its mathematical scope

Quantity	Construction checker	Independent checker
Working controls / transition rows	194 / 388	194 / 388
Primitive source sites	3,650	3,650
Source outcomes	4,750	4,750
Symbolic steps	1,514,674	1,101,153
Maximum symbolic steps per outcome	508	403
Exact blank startup	57,768	57,768

The independent checker also records a minimum positive-duration constant of 3, four generic list lemmas rederived from literal rows, and exact agreement of two source-graph resolvers. The separate simultaneous fold check has 32 cases. Symbolic-step counts measure verification work, not the machine's running time on arbitrary register values.

The construction checker starts from the frozen source tree and physical compiler. It rebuilds the 202-state raw machine, checks the eight simultaneous folds, checks every source outcome, and compares the exact blank-start tape with its intended seed. The independent checker imports neither that compiler nor its arithmetic-box oracle. It reconstructs the new table from the parent table and row (2), reconstructs the seven-site source replacement by binary addresses, checks two separately implemented control-flow resolvers, and proves its four generic register-list lemmas from the literal transition rows before checking all source outcomes.

4.1 What makes the symbolic checks universal

A represented natural value v has block $B(v) = 0\,1^{v+1}$. The logical-to-physical register map is

$$(6, 2, 0, 4, 3, 7, 8, 5, 1),$$

for logical registers $(N, T, W, A, D, H, X, Y, L)$. The symbolic frame contains nine independent arbitrary natural register values, the first arbitrary spare value, an arbitrary finite remaining spare list, a header distance $48 + n$ with $n \geq 0$, and blank exterior. At source address p the intervening zero gap is $48 + n - |p|$. For a decrement the zero case and positive case $1 + v$ are separate contracts; neither is omitted because a finite trace fails to visit it.

The certificate rules are literal transitions, constant-symbol scan induction, affine arithmetic on run lengths, and induction over a finite list of represented register blocks. The independent checker rederives the one-block scan identities with arbitrary exterior contexts. A block of value v takes $v + 2$ steps for each relevant traversal, and induction extends the result to a list. For the alternative block layout, put $C(v) = 1\,0\,1^v$ and extend B and C to finite lists by concatenation. The identity $C(A)1 = 1B(A)$ follows from the empty-list and one-block identities. These rules support arbitrary values rather than a bounded sample of register assignments.

Every closed source contract has a strictly positive physical duration. The documented duration is an affine expression in nonnegative frame parameters with positive constant term. Thus compressed list operations cannot create a spurious zero-step source simulation. For increments, the expected frame includes the appended spare block; for positive decrements it includes the changed header distance. Exact complete-frame equality, including head movement and untouched exterior, is required.

The proof status is ordinary mathematical verification using these sound symbolic rules and executable certificates. The Python engines and their induction rules remain part of the trusted verification base. A successful run of an arbitrary Python program would not establish a PA theorem; the elementary rule interpretations and the PA-internal trace argument explain why these specific certificates have that role.

4.2 Blank initialization is checked separately

Both checkers independently run the literal machine from the all-zero tape for 57,768 transitions. The result is the exact source-25 frame at native state 84 and head -2, with logical vector $(2, 0, 0, 0, 0, 0, 0, 0, 0)$ and 26 spare blocks of value one. The first uses a finite-support set; the second uses a byte array and checks every cell of the array while checking that the head never leaves it. The unvisited exterior is therefore still zero. Equality is not limited to the head and state or to a guessed local window.

This finite startup check proves a finite binding fact. The infinite nonhalting equivalence comes from the compiler, annotation invariant, universal source-step simulation, and PA calibration. No finite-duration simulation is extrapolated into an infinite-run assertion.

5 The inherited logical calibration, made explicit

The numerical formula constructors remain

$$I(a, b) = 4p(a, b), \quad M(a, b) = 4p(a, b) + 1, \quad U(a, B) = 4p(B, a) + 2, \quad E(a, b) = 4p(a, b) + 3.$$

Code zero denotes truth, one is $M(0, 0)$, and negation is $I(B, 1)$. Every variable label occurring in formula code B is strictly less than B, with zero handled separately. Hence $\Gamma(B) = B \to U(B, B)$ is a

valid freshness/vacuity instance. The separate-head source emits this head and the guarded separation gate separately, together with the retained six logical heads and the dominated implicational head.

For completeness, the six universally instantiated heads are

$$
\begin{aligned}
&(\exists x_a\, U(a, C)) \to C,\\
&U(a, U(b, C)) \to U(b, U(a, C)),\\
&M(c, a) \to (E(c, b) \to M(b, a)),\\
&E(c, a) \to (M(b, c) \to M(b, a)),\\
&\exists x_a E(c, a),\\
&U(a, b) \to \big(U(a, b \to c) \to U(a, c)\big).
\end{aligned}
$$

Existential quantification and conjunction here use their classical implicational abbreviations. The dominated head is $((H \to d) \to D) \to ((D \to H) \to (A \to H))$ with $d \le D$. The guarded head is $G(a, b, c) = \neg(V(a, b, c) \to Q(a, b, c))$ when $c \neq a, b$, and truth otherwise, where

$$
\begin{aligned}
V(a, b, c) &= M(a, b) \to U(c, M(a, b)),\\
P(a, b, c) &= c \to M(c, b),\\
S(a, b, c) &= U\big(b, (M(a, b) \to P(a, b, c)) \wedge (M(c, b) \to c)\big),\\
Q(a, b, c) &= U(c, S(a, b, c) \to \neg P(a, b, c)).
\end{aligned}
$$

Truth, these heads, and the separate $\Gamma(B)$ instances generate the numerical calculus under modus ponens and generalization of theorems. The compiler's AX parameters instantiate this gate as $G(H, D, A)$. The free-term emitter check binds this complete ordered family to the frozen 190-token source word [2, 3].

For the direction from a source halt to a PA refutation, interpret $M(a, b)$ as $\mathrm{Bit}(x_a, x_b)$, $E(a, b)$ as equality, and U as universal quantification. PA proves finite binary comprehension for each arithmetic formula. Crucially, its proof is a uniform syntactic template: given a formula code, produce the corresponding induction instance and the finite PA derivation. PA verifies the resulting proof code. This does not assert a uniform truth predicate or a reflection principle for PA.

The inherited proof [5, 2] separately handles coincident labels in the separation gate, so bound occurrences are not incorrectly treated as free. It also handles $I(0, 0) = 0$ by an explicit truth equivalence and canonical Cantor-stack collapse by the literal pairing identities. Every visible formula and live canonical stack cell carries a PA proof annotation. A halting Shift therefore extracts a PA proof of code one's translation and then a PA proof of $0 = 1$.

For the opposite direction, the PA196 refutation compiler is retained through the diagonal-calculus replacement of its equality interfaces [6, 5, 2]. The supplied proof derives reverse membership rewriting using a fresh label equal to a formula's own code and the diagonal head, then derives the required neighborhood-equality implication and witness. These replace the old equality-vacuity interface in the compiler. The separate-head version has nine generated stack entries plus the saved top, so its selectors must use the ten-entry protocol. Our command proof checks that protocol again for PA194NS.

The audit in `comparison/calibration_audit.txt` found substantive proof-producing arguments for these steps and no immediate defect in the inspected interfaces. This is a review of the retained ordinary proofs, not a new formalization of the PA196 compiler or of all its transitive dependencies. Complete relevant inherited proof texts and their provenance are included in `references/logical/`.

6 Search below 194 and the limitations of the negative result

Heuristic run	Proposals	Best effective AX cost
run1	9,027,164	60
run2	8,332,404	60

The two recorded decoder runs made 17,359,568 proposals in total and found no cost below 60. The final crossing screen also tested 20 fixed grammar/closing/entry combinations; every optimistic count in its local-fold template was 194. Neither count is a theorem about all possible machines.

Physical search	Radius	Step bound	Candidates
Two-control merge, all-tape windows	4	100	0
Two-control merge, source/startup windows	5	150	0
Incoming-edge elimination, source/startup windows	5	150	0

The refined input has 16,293 conservative windows for 171 non-box controls, collected from all 4,750 source outcomes and exact startup. Arithmetic-box controls retain the broader abstraction. These are bounded negative results only.

The decoder search starts from the frozen 67-reader, 68-reader, and quantifier-head-exchanged shapes, using PA194NS's actual fixed-controller rows. It jointly changes shared and occurrence-level binary associations, the order of the seven freely emitted heads, and parameter permutations. The objective is the number of AX readers minus the maximum compatible history folds, with explicit scanner, address-depth, and structured-leaf constraints. With 134 fixed controls after the controller fold, a valid effective AX cost of 59 would yield 193. The best observed cost remains 60. These are proposal counts, including repetitions and rejected mutations, not counts of distinct certified machines.

Separate physical searches try to replace two controls by one and to eliminate a control by redirecting its incoming edges. They compare exact future local configurations within stated window and time bounds, first with arbitrary-tape windows and then with conservative windows gathered from every symbolic source contract and complete startup. The supplied receipts state each window radius, time bound, and context abstraction; none supplies a new contraction.

The short-scanner idea was examined directly. Removing a detector state would require a stronger global address invariant. The first legal source address already has six consecutive zero bits, and a direct deletion fails. Rotating the shared comparison subtree can shorten that path but changes structured break destinations, so flattened-word equality is insufficient. Transient increment frames and blank initialization must also satisfy any shortened-scanner proof. No accepted reduction results from these experiments.

An independently constructed top-only Shift alternative passes the construction checker at 195 states; the extra transfer reader costs a control. It is retained as a research result rather than advertised as a smaller machine. The accepted 194-state design is the explicitly verified PA194NS table. None of these bounded or heuristic failures establishes that 193 states are impossible.

7 Relation to ZFC and concise-iPRA work

The ZFC zero-spare-transfer report already develops no-push MP and the idea that the proof-word compiler can reserve the zero formerly inserted by the physical machine [7]. Its Shift tests differ, so its theorem cannot simply be cited for PA194NS. The useful transferable idea is the strong framed

contract, valid from every top and every underlying stack, with earlier halting allowed. The present proof rebuilds that contract for the ten-entry PA emitter and the cache-only Shift test, including the new final detector.

The concise-iPRA project compares complete proof checkers and proves translations between presentations [8]. Its transferable lesson here is to count and verify the complete interface: removing one action from a checker moves an obligation to the proof input or to a derived rule. PA194NS moves the spare-zero obligation to its proof compiler. We do not replace PA by iPRA or infer PA consistency from a smaller iPRA checker. The target remains the same PA refutation predicate, and Python byte count is not used as a proxy for binary TM state count.

8 Reproduction and package guide

Python 3.10 or later and its standard library suffice for the accepted construction and verification. Keep assertions enabled; do not use `python-O` or `PYTHONOPTIMIZE`. From the package root:

```
python3 -B verify_all.py --quick
python3 -B verify_all.py
```

The quick mode checks integrity, exact tables, local folds, command-word identities, complete startup, and graph novelty. It deliberately omits the two long symbolic physical replays. The default performs those replays as well. Fresh receipts are written to a temporary directory, not over the delivered evidence. Individual commands are listed in the top-level README and permit a user-selected evidence directory.

Path	Content
`machine/`	Dense and native tables, complete source and source-address maps, label bijection, eight folds, seed and construction receipt.
`construction/`	Deterministic source/tree compiler, row edit, simultaneous fold checks, symbolic physical checker, and frozen references.
`verification/`	Independent address splice, two control-flow resolvers, literal-table symbolic engine and complete recorded evidence.
`proof/`	The new command proof and executable free-constructor/command-word checks.
`comparison/`	All-packet ledger, all 194-state classes, full census, exact old tables, novelty checker and calibration audit.
`references/logical/`	Inherited ordinary PA-calibration arguments and attribution.
`research/`	Optional bounded searches, frozen inputs, receipts and unsuccessful constructions; no additional successful smaller-state claim.
`report/`	This report, complete local LaTeX inputs, generated tables and PDF.

The optional searches require a C++17 compiler. They are not dependencies of the accepted machine: the frozen grammar, literal table and certificates define the construction. The full census can be regenerated from the pinned ProveIt checkout using the included scanner; graph novelty can be replayed offline against the retained old 194-state tables.

To rebuild the report, run twice from `report/`:

```
pdflatex -interaction=nonstopmode -halt-on-error pa194ns_report.tex
```

The release manifest identifies every delivered file except itself. Recompiling the PDF can change metadata and consequently its file hash; verify the supplied manifest before regenerating artifacts.

A Complete transition table

Each entry is wDq: write bit w, direction $D \in \{L, R\}$, target q. H denotes the terminal label 194. The start is control zero on the blank zero tape. The table below is generated from the authoritative dense file.

State	Read 0	Read 1	State	Read 0	Read 1
0	1R1	1R2	97	0R112	1R89
1	1R3	1R1	98	0R112	1R104
2	0R0	1R2	99	0R113	0L14
3	0R4	0R1	100	0R114	1L66
4	1R5	1L6	101	0R115	1R116
5	1L7	1L8	102	0R117	1R102
6	0L7	1L6	103	0R14	1R103
7	0L9	1L6	104	0L109	1L33
8	0L10	0L11	105	0R67	1R118
9	1R12	0L9	106	0R119	1L120
10	0L9	0L10	107	0R121	1R122
11	1L8	1L11	108	0R46	0L14
12	0R12	0R13	109	0R85	1R94
13	0L14	0R13	110	0R109	1L89
14	1L15	0L15	111	0R123	1R123
15	0L15	1L16	112	0L60	0R82
16	0L17	1L16	113	0R62	0R117
17	0L18	1L16	114	0R55	0R117
18	0L19	1L16	115	0L80	1R124
19	0L20	1L16	116	0R125	1R126
20	0L21	1L16	117	0R127	1R117
21	0L22	1L16	118	0R47	1R84
22	0L23	1L16	119	0R55	0R127
23	0R23	1R24	120	0R35	1R33
24	0R25	1R26	121	0R93	1R108
25	0R27	1R28	122	0R74	1R73
26	0R29	1R30	123	0R46	1R46
27	0R31	1RH	124	0R128	1R129
28	0L32	1R33	125	0R32	1R64
29	0R34	1R35	126	0R31	1R92
30	0L33	1L36	127	0R63	1R127
31	0R37	1R38	128	0R130	1R131
32	0R39	1R40	129	0R132	1R133
33	0R41	1R42	130	0R134	1R135
34	0R43	1R44	131	0R136	1R137
35	0R45	1R36	132	0R138	1R139
36	0R46	1R47	133	0R140	1R141
37	0R48	1R41	134	0R91	1R120
38	0R49	1R50	135	0R142	1R123
39	0R51	1R52	136	0R143	1R144
40	0R53	1R54	137	0R145	1R146
41	0R55	0R56	138	0R134	1R147
42	0R57	0L14	139	0R46	1R148
43	0R58	1R59	140	0R149	1L131
44	0R60	1R49	141	0R150	1R151
45	0R55	0R61	142	0R71	1R40
46	0R62	0R63	143	0R28	1R152

State	Read 0	Read 1	State	Read 0	Read 1
47	0L64	1L65	144	0L153	1L154
48	0R45	1R50	145	0L129	1R155
49	0L66	0L15	146	0R156	1R157
50	0R67	0L10	147	0R149	1R158
51	0R68	1R69	148	0R159	1R46
52	0R70	1R71	149	0R91	1R160
53	0R72	1R73	150	0R161	1R162
54	0R74	1R75	151	0R163	0L10
55	0L76	0R1	152	0R164	1R165
56	0R61	1R56	153	0R166	1R157
57	0L68	1L35	154	0R167	1R156
58	0R77	1R78	155	0R135	1R130
59	0R51	1R79	156	0R147	1R40
60	0L80	1R81	157	0R134	1R168
61	0R82	1R61	158	0R40	1L169
62	0L76	0R2	159	0L170	1R171
63	0R83	1R63	160	0R100	1R79
64	0R30	1R84	161	0R172	1R170
65	0R85	1R86	162	0R173	1R174
66	0R35	1R87	163	0R175	1R176
67	0R41	0L13	164	0R177	1R178
68	0R88	1R65	165	0L179	1L180
69	0R47	1R89	166	0R149	1R140
70	0R41	1R90	167	0R165	1R181
71	0R91	1R92	168	0L130	1R46
72	0R93	1L94	169	0R182	1R135
73	0R47	1R72	170	0R183	1R145
74	0R87	1R95	171	0R158	1R130
75	0R47	1R96	172	0R139	1R165
76	1R76	1L15	173	0R158	1R184
77	0R88	0L7	174	0R185	1R100
78	0R97	1R98	175	0R31	1R49
79	0R41	1R99	176	0R186	1R175
80	0R100	1R84	177	0R134	1R142
81	0R101	0L67	178	0R32	1R187
82	0R102	1R82	179	0R144	1R188
83	0R103	1R83	180	0R189	1R190
84	0R104	0L14	181	0R40	1R191
85	0R62	0R102	182	0R168	1R40
86	0R105	0L84	183	0R192	1R46
87	0R30	1R57	184	0R153	1R146
88	0L60	0R83	185	0R165	1R180
89	0R88	1L106	186	0R118	1R106
90	0R95	0L14	187	0R193	1R159
91	0R64	1R107	188	0R169	1R154
92	0R106	1R70	189	0R158	1R192
93	0L60	0R103	190	0R179	1R182
94	0L59	1R108	191	0R182	1R40
95	0L109	1R110	192	0R149	1R168
96	0R111	0L14	193	0R71	1R144

B All existing 194-state graph classes

The role labels report the status supplied with each old design. Historical certificates were compared and inspected, not all rerun. Exact identities, all occurrences and the associated source paths are in the CSV/JSON files.

Graph representative	Role	Difference and evidence boundary
M4R `a3c1d556d61b`	Principal report	Coefficient-4 source graph unchanged, new decoder arrangement; 135 + 67 - 8 = 194.
M5 `180a0fd1f046`	Principal report	Formula-constructor coefficient 5 with new uniform proof recoding and generic cons proof; 135 + 67 - 8 = 194.
M6 `725ce8474d7b`	Principal report	Formula-constructor coefficient 6 with new uniform proof recoding and generic cons proof; 135 + 67 - 8 = 194.
PA194-67 `4bfea62b17aa`	Principal report	Separate diagonal head; 67 AX readers and seven compatible history folds.
PA194-68 `9087d70fc0e1`	Principal report	Different association of the same separate-head emitter, with eight folds.
PA194-AD `27613aa92da7`	Principal report	Exchange the physical A and D register positions; three changed native rows.
PA194A `3c3e74ffcb5d`	Principal report	PA194CS entry jump at source 26 becomes decrement A; finite zero invariant and two symbolic physical checks.
PA194C `d759313c03b5`	Principal report	Cached Shift/MP controller on PA194-67; six changed native rows. All five later cached packets duplicate this exact dense table.
PA194CS `1c71002e5b8a`	Principal report	Destructive equality operand symmetry shortens Shift by three sites. Extra controller reader 330 is folded with 41; this is not an extra AX fold.
PA194CSZ = PA194X `62e93098caf7`	Principal report	Symmetric cached Shift plus known-zero X decrement at source 26. PA194X and PA194CSZ have identical dense bytes.
PA194D `4226b7702de5`	Principal report	Separate diagonal clause Gamma(D) replaces Gamma(A); new selector and 134 + 69 - 9 allocation.
PA194F `4fcee0c2ef7b`	Principal report	PA194CS final emitter jump becomes decrement X; four-site zero invariant and two physical checkers.
PA194J `7c498ced0eb0`	Principal report	Ten jumps replaced by proved-zero decrements, affecting three native rows; 263180-pair positivity invariant.
PA194N `37f7e909724c`	Principal report	Successful MP returns its consequent in T and omits the dummy-zero push; original Shift t+h=1 remains.
PA194Q `48e98b5504dc`	Principal report	Exchanges P and Q in the curried quantifier-distribution axiom; 79 source instructions change, with explicit exchange proof.
PA194R `8a9a8a30773f`	Principal report	PA194F source with 1111 changed address paths; 135 fixed + 67 AX readers - 8 folds = 194.
CS both-zero6 seed `28b3a48e3485`	Research seed	Additional known-zero-jump combination retained as a state-search input; no separate principal consistency endpoint.
Cached I-tree zero-jump seed `1dd4bf6a2c6b`	Research seed	Cached I-tree zero-jump input for scanner permutations and closing-register experiments; not a principal output.
Gamma(D)-CSZA `f9527c1867a0`	Supplement	Additional Gamma(D)/symmetric-Shift/zero-jump sibling with primary-checker evidence; no separate independent physical replay.
MPH seed `f668a2f4de78`	Research seed	MP result goes to H while Shift is original; supplied as an additional physical merger/bypass search kernel.

Graph representative	Role	Difference and evidence boundary
PA194CSZ_A3 `57a1edbaafb0`	Supplement	Zero-jump sibling choosing final register 3; primary-checker/invariant/startup evidence, no separate independent physical replay.
PA194CSZ_A5 `2bcebbeefd95`	Supplement	Zero-jump sibling choosing final register 5; primary-checker/invariant/startup evidence, no separate independent physical replay.
PA194CSZ_A6 `4bccc41f2e84`	Supplement	Zero-jump sibling choosing final register 6; primary-checker/invariant/startup evidence, no separate independent physical replay.
PA194CZ `b713e73457be`	Supplement	Replace source-site-26 jump by a zero test of register six; native row (27,1) changes from 0 L 40 to 1 R 21.
PA194altC `a509952c1e06`	Supplement	Cached controller composed with the alternate PA194-68 reader allocation.
Rejected CS node1-X `55ef6a817807`	Rejected	Source invariant and source contracts pass, but the literal blank machine halts after 1361 steps before its clean seed.
Rejected I-tree node1-X `f8abcd947786`	Rejected	Known-zero logical-source change fails blank initialization; documented halt at 1361 literal steps.
Rejected scanner194 `c48d6711e9c7`	Rejected	Optimistic shortened-scanner/reader quotient fails the required source simulation.

C Packet-by-packet ledger

The ledger covers the 41 filed directories and five incoming archives at the pinned revision. A negative result is recorded where the packet reports one; it is not promoted into a global bound. Archive and README hashes are retained in the machine-readable ledger.

Packet	Reported contribution and role
1. Below_U208_Research_2026-10-29	Derives sound seven-cell source-boundary guards and a head-offset quotient lemma for physical reduction searches. Whole-loop pruning removes five source sites but increases the decoder DAG from 197 to 200 nodes; offset, neutral-row and two-row searches produce no accepted U207.
2. ConPA_below208_research_2026-10-29	A two-element countermodel rejects identifying the cover source and output. Exhausting 279,936 independent parameter permutations finds minimum decoder DAG 197; the reported two-row synthesis searches also produce no accepted U207.
3. PA195_ConPA_research_2026-11-14	Introduces a 67-reader allocation for the separate 190-token emitter, producing PA195S with 134+67-6 states. Its Combined-controller sibling PA195C has 3,649 sites; both have two independent physical checks and exact blank startup.
4. PA_consistency_TM_research_2026-11-13	Combines a shorter negation and MP source to reach 3,573 sites and a 157-node decoder, and finds a five-zero-address layout of the original source. It adapts the ZFC opposite-comparison residual to parameter-free MP with a proof compiler, but does not realize a smaller accepted physical table.
5. U208_ConPA_2026-10-29	Reduces Joint U209 by deleting state 211 in favor of 172, using a three-cell fold with positive blocks of lengths 3, 1, 7 and 1. This is the same saving under the opposite representative choice, so it cannot be added to the Conditional Context saving.
6. U208_ConPA_Conditional_Context_2026-10-29	Reduces Joint U209 to U208 by deleting native state 172 and redirecting the two incoming controls through 211. A five-state, three-cell history invariant gives positive simulation blocks and all-tape equivalence; all 5,084 source contracts are checked.
7. U208_ConPA_Entry_Context_2026-10-29	Independently justifies the same literal U208 table as the Conditional Context packet using depth-two and depth-three entry histories, with corruption controls. A compatibility helper yields a 214-state companion, not a further state saving.
8. U208_ConPA_History_2026-10-29	Reduces the Plain U209 layout by deleting state 211 in favor of 172 and altering the zero transition to state 214. The five-role history relation explains its equivalence to the alternate reduction; a source-fold proposal is rejected for routing to successor 44 instead of 28.
9. conpa_tm_comparison_and_reduction_search	Proposes two one-row MP variants at 196 states with command-level arguments but only bounded physical evidence. Two Shift variants are rejected because their blank runs halt after 43,563 steps; the packet distinguishes a proved one-way endpoint from a reverse proof schema.

Packet	Reported contribution and role
10. conpa_tm_comparison_and_search	Swaps the physical A and D registers to obtain a distinct 196-state table and a 58,122-step startup. A two-rotation 195-state proposal passes the source CFG check but fails 93 physical contracts; reflexivity replacing the witness head also has an insufficient-strength Boolean model.
11. conpa_tm_study_2026-11-13	Introduces the Combined controller: H-only Shift and successful MP returning in H, yielding a 196-state machine with 3,788 sites and 4,928 outcomes. Direct negation yields 197 states; a 157-node source realization needs 207 physical states, and the parity component remains incomplete.
12. pa194C_conpa_2026-11-13_2	Supplies the cached PA194C composition, changing six native rows and reducing the source from 3,673 to 3,649 sites. The packet includes independent verification of all 4,749 outcomes and 188 command-algebra checks; its dense table matches the other PA194C packets exactly.
13. pa194_conpa_2026-11-13	Combines the best 67-reader allocation with seven compatible folds to obtain the first principal 194-state separate-head table, with 3,673 sites and 4,780 outcomes. It includes an A/D physical-register companion and independent literal-tape verification.
14. pa194_conpa_2026-11-13_2	Obtains another 194-state decoder by the alternative 134+68-8 allocation for the same separate-head word. The eight folds have 32 local cases, and the report supplies ten-entry stack accounting plus independent physical and startup checks.
15. pa194_new_designs_2026-11-13	Supplies PA194C and the distinct PA194CS symmetric destructive-comparison Shift, cutting three more source sites. PA194CS offsets an extra reader by the controller fold 41 to 330, and verifies all 4,745 outcomes plus its 57,768-step startup.
16. pa194c_conpa	Independently supplies the same PA194C table and a cached-controller companion using the alternative 68-reader/eight-fold allocation. The companion has primary-checker evidence without a separate independent replay; scanner-macro searches do not improve the state count.
17. pa194c_conpa_2026-11-13	Independently verifies the same PA194C table and reconciles 34 earlier archives. Deleting scanner state 34 fails 113 contracts and 698 two-row routings fail; 1,308 interface-changing candidates remain unchecked, so none is a verified 193-state result.
18. pa194c_conpa_2026-11-13_1	Supplies the same PA194C and supplementary PA194CZ, which replaces the site-26 jump by a register-6 decrement. A 263,244-pair zero invariant supports the CZ physical interface and 58,168-step startup, but that sibling has no separate independent checker.

Packet	Reported contribution and role
19. pa195_conpa_2026-11-13	Obtains 195 states from the bundled 191-token diagonal emitter by retaining seven compatible folds: 134+68-7. The 3,700-site source has 4,815 verified outcomes, and the packet gives detailed uniform PA calibration and an all-tape proof of the extra fold.
20. pa195_diagonal_combined_2026-11-13	Composes the bundled diagonal emitter with Combined command semantics, changing six native rows and yielding PA195DC at 134+68-7 states. Its 3,676-site source has 4,784 outcomes and a 57,922-step startup; optional negative-diagonal ideas lie outside the endpoint.
21. pa195_followup_research	Builds valid M195S8 by adding scanner control that makes eight folds usable, retaining 195 states. Its purported 194-state seven-zero-address layout is rejected by full physical contracts despite a valid source CFG and local fold proof; Gamma relocation and gate bundling remain research directions.
22. pa196_research_2026-11-13	Finds a 157-node source decoder, down from 159, while preserving the 3,812-site control flow. The apparent 195-state count loses an existing decoder-helper fold; the native HA calculus and stack-repair study do not supply an accepted smaller literal machine.
23. pa208_research_2026-10-29	Supplies 18 graph-distinct 208-state helper variants, each with a positive two-cell local proof. Directional reassociation and shorter negation reduce source cost but produce 212 states; reserved-zero compilation for no-push MP produces 209, illustrating the gap between source and physical cost.
24. pa_below196_research	Compares eight equivalent gate spellings and proves a six-implication minimum within a restricted expression search. Constructor transposition, pairing compression and a seven-site H-only Shift reduce selected source costs but worsen other decoder or guard costs; none yields a smaller physical machine.
25. pa_consistency_comparison_2026_11_13	Proves that the first quantifier head is equivalent to modal B plus explosion, with explicit Hilbert certificates, and tests further heads with countermodels. Twelve source combinations and a 189-token joint emitter leave scratch-register and compiler interfaces unresolved, so no smaller physical endpoint is established.
26. pa_consistency_comparison_and_new196_2026-11-13	Supplies three complete 196-state diagonal/separate-head layouts, including the 190-token, 3,673-site separate emitter. A hybrid U208/PA196 kernel passes source contracts but every tested start state escapes on blank tape, while the proposed 195-state short scanner fails 96 contracts.
27. pa_consistency_cross_archive	Builds four 196-state compositions of diagonal/separate emitters, Combined command semantics and a proved-zero entry jump. It supplies explicit nine- versus ten-entry compiler and protected-tail accounting; the shorter-negation composition remains at 197 states.

Packet	Reported contribution and role
28. pa_consistency_reduction	Provides a rebracketed 196-state decoder and a 196-state H-only Shift variant. Integrating the equality-witness gate saves 67 source sites but requires 206, then 203, physical states; a nominal warm-start 195-state candidate is trapped in a four-step cycle.
29. pa_consistency_reduction_audit_2026-11-14	Builds four MP-in-H compositions with diagonal/separate emitters and optional short negation, at 196, 196, 197 and 197 states. Their physical and startup checks pass, but the packet explicitly leaves logical calibration, including the scratch-zero interface, uncertified.
30. pa_consistency_research	Develops geometry-safe 158-node reassociations; an alternative 158-node layout still needs 196 states because its eight-zero addresses consume the expected fold saving. It verifies 4,959 universal contracts and a 59,164-step startup, and uses countermodels to test membership-rewrite simplifications.
31. pa_consistency_research_2026-10-29	Proves a 3,909-to-3,904-site specialization using inductive zero invariants for registers 1 and 6, without reducing the physical state count. It also reconciles both U208 graph classes and diagnoses a failed 207-state rotation repair with exact interface guards.
32. pa_consistency_research_2026-11-13	Replaces equality vacuity by diagonal generalization, reducing the axiom word from 201 to 191 tokens and the source from 3,812 to 3,700 sites. Its first implementation has 199 states, and its report supplies the uniform PA finite-bit-comprehension soundness argument used by later packets.
33. pa_consistency_research_2026-11-13_2	Verifies twelve 196-state reader rotations and reduces the diagonal emitter from 199 to 198 states by an additional fold. It also studies halting directly on successful MP and a large joint-register abstraction, without a smaller complete blank-input table.
34. pa_consistency_research_2026-11-14	Uses p(t,1)=p(0,t)+2 to replace eight negation expansions, cutting 3,673 source sites to 3,441. The resulting table has 197 states because only five folds remain; a 245,112-pair invariant supports all 4,468 physical outcomes and blank startup.
35. pa_tm_comparison_and_reductions	Provides two rotated 196-state layouts and a distributed-gate emitter that reduces 201 to 199 tokens and 3,812 to 3,773 sites but uses 198 states. A 67-rule grammar remains a source-level result, while a core countermodel rules out a tempting factor movement.
36. pa_tm_extended_research_2026-11-13	Supplies four 196-state variants replacing a closing jump by a proved-zero decrement, each with all 4,959 constrained contracts and blank startup. Refined joint register invariants remove five reachable outcomes; a quantifier-head replacement is rejected by a countervaluation.

Packet	Reported contribution and role
37. pa_tm_research_2026-11-13	Reorders the six emitted heads to shorten maximum decoder path from 38 to 36 while retaining 196 states; all 4,959 source contracts and the 58,154-step startup pass. A fresh Lean build establishes physical/source equivalence, not a complete arithmetic consistency calibration.
38. pa_tm_research_2026-11-13_2	Uses a proved-zero register-6 decrement at source site 26 to obtain another 196-state layout with startup 58,402. Of 936 jump-action products, 671 pass constrained contracts, but that does not certify every startup; an attempted 195-state fold has an actual blank-run error.
39. pa_unary_power_cover	Introduces an equality-free one-relation finite unary-power-cover theory and proves its mutual interpretation with full PA. Gives a numerical emitter with nine nonlinear multiplications and a 20-node wrapper; it does not itself supply a new literal TM.
40. u208_comparison_research_2026-10-29	Classifies the two U208 graph families and rejects a 207-state candidate with an exact first-return counterexample. It proves equivalence of weak and exact logical schemas, but leaves the costed theory replacement's PA, equality and runtime interfaces unresolved.
41. u208_research_2026-10-29	Reconciles the two U208 action-graph classes and verifies all 5,084 primitive contracts for both literal tables. It records the logical and formalization dependencies of the consistency endpoint; the repeated tables do not constitute additional independent reductions.
42. PA194R_research_2026-11-14	PA194R reassociates the PA194CS decoder and replaces a known-zero final jump; PA194F is its one-row intermediate, PA194A and PA194X alter the entry jump. PA194X duplicates PA194CSZ. Rejected native-103 change halts after 1361 steps.
43. pa194N_conpa_2026-11-13	PA194N removes the successful-MP dummy-zero push, retains MP result in T and the original t+h=1 Shift. New ten-entry command compiler; 3654 source sites. MP-to-H/original-Shift table is only a physical-search seed.
44. pa194_radix_family_2026-11-13	M5 and M6 change formula-constructor coefficients to 5 and 6 with proof-code recoding. M4R preserves coefficient 4 but changes the decoder. All three have 194 states. Includes PA194CSZ, additional research seeds and a rejected startup change.
45. pa194j_pa194d_conpa_2026-11-13	PA194J replaces ten source jumps by proved-zero decrements. PA194D replaces separate Gamma(A) by Gamma(D), with 69 AX readers and nine folds. Additional final-register and Gamma(D) siblings are supplementary primary-checker results.
46. pa194q_conpa_2026-11-13	PA194Q exchanges curried premises in the quantifier-distribution head, with a bidirectional K/S/MP proof translation. PA194CSZ combines symmetric Shift and a proved-zero entry jump; it is the same literal dense table as PA194X.

References

[1] ProveIt, *PA consistency machines*, pinned revision `1e407ae841e4666f932777fad977ba784ebbcfd5`, https://github.com/VladimirReshetnikov/ProveIt/blob/1e407ae841e4666f932777fad977ba784ebbcfd5/Computability/BusyBeaver/Research/PA/README.md.

[2] ProveIt, *A 194-state Turing machine calibrated to Con(PA)*, filed research packet `PA/docs/pa194_conpa_2026-11-13`, especially `report/calibration.tex` and the source/physical certificates.

[3] ProveIt, incoming `pa194N_conpa_2026-11-13.zip`, especially `mathematics/full_nopush_proof.tex` and the frozen command and physical verifiers.

[4] ProveIt, incoming `PA194R_research_2026-11-14.zip`, especially the PA194CS inputs, symmetric-Shift proof and literal symbolic runtime used as the physical construction basis.

[5] ProveIt, original PA195 calibration, `PA/docs/pa195_conpa_2026-11-13/report/sections/pa_calibration.tex`; the separate-head and diagonal refinements are retained under `references/logical/` in this release.

[6] James Chang, imported PA196 and PA209 developments under `PA/jamesyc-upstream/`, including `PA196/Logic`, the source compiler, and the explicit PA/iPRA proof-status statements.

[7] ProveIt, *Zero-spare transfer*, section "Removing the successful MP zero push", `Computability/BusyBeaver/Research/ZFC/candidate-k215/alternatives/zero-spare-transfer/report.tex`.

[8] ProveIt, *Concise presentations of intuitionistic primitive recursive arithmetic*, https://github.com/VladimirReshetnikov/ProveIt/blob/1e407ae841e4666f932777fad977ba784ebbcfd5/Logic/PrimitiveRecursiveArithmetic/ConciseIPRA/README.md.